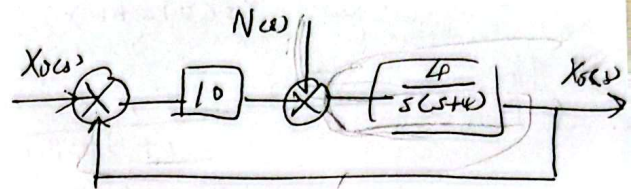


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HW

Q1 稳态误差 $(e_{ss}) = ?$

$\therefore$ 单位反馈
 $H(s) = 1$



$$\frac{E_N(s)}{N(s)} = \frac{G}{1 + GH} \quad \therefore e_{ss} = e_{ss}$$

$$|H| = 1, |G| = \frac{4}{s(s+4)} \cdot 1, \quad \therefore n(s) = -2.1(t) \\ N(s) = \frac{-2}{s}$$

$$\therefore E_N(s) = \frac{\left(\frac{4}{s(s+4)} \cdot 1\right)}{1 + \left(\frac{4}{s(s+4)} \cdot 1\right) \cdot 10} \cdot \left(-\frac{2}{s}\right)$$

$$\frac{G}{1 + GH}$$

$$G = \frac{4}{s(s+4)}$$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s E_N(s) = \lim_{s \rightarrow 0} \frac{-4}{s^2 + 4s + 40} (-2) = +0.2$$

$$\frac{E_{X_0}(s)}{X_0(s)} = \frac{1}{1 + GH}$$

$$G = 10 \cdot \frac{4}{s(s+4)}$$

$$X_0(t) = t \cdot 1(t)$$

$$X_0(s) = \frac{1}{s^2}$$

$$n(t) = -2.1(t)$$

$$N(s) = \frac{-2}{s}$$

$$\therefore E_{X_0}(s) = \frac{1}{1 + 10 \cdot \frac{4}{s(s+4)}} \left(\frac{1}{s^2}\right)$$

$$e_{ss} = \lim_{s \rightarrow 0} s E_{X_0}(s) = \lim_{s \rightarrow 0} \frac{s(s+4)}{s^2 + 4s + 40} \left(\frac{1}{s}\right) = +0.1$$

$$\therefore e_{ss} = e_{ss_N} + e_{ss_{X_0}} = 0.3$$

④ 单位反馈

Q2

$$G(s) = \frac{100}{s(0.1s+1)}, \quad x_0(t) = (1+t+at^2) \cdot 1(t), \quad a \geq 0$$

解 a

$$\begin{aligned} \frac{E(s)}{X_0(s)} &= \frac{1}{1+G(s)H(s)}, \quad H(s)=1, \quad \therefore e_{ss} = E_{ss} \\ &= \frac{1}{1+G(s)} \\ &= \frac{s(0.1s+1)}{0.1s^2+s+100} \end{aligned}$$

$$x_0(t) = (1+t+at^2) \cdot 1(t)$$

$$X_0(s) = \frac{1}{s} + \frac{1}{s^2} + \frac{2a}{s^3} = \frac{s^2+s+2a}{s^3}$$

$$E(s) = \left(\frac{E(s)}{X_0(s)} \right) \cdot X_0(s)$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \frac{E(s)}{X_0(s)} \cdot X_0(s)$$

$$\therefore \begin{cases} e_{ss} = \infty & (a > 0) \\ e_{ss} = 0.1 & (a = 0) \end{cases}$$

Q4

$$\theta_d(t) = [10^\circ + 60^\circ t (1+t)] \cdot 1(t)$$

$$e(t) = \theta_d(t) - \theta_o(t)$$

$$G(s) = \frac{400(0.3s+1)}{s^2(s+1)}, \quad k=2 \text{ (II-)}$$

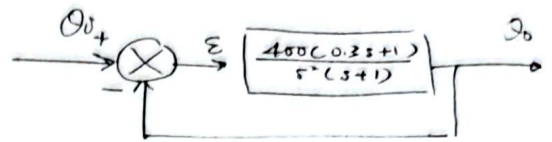
$$e_{ss} = \frac{A}{1+k_p} + \frac{B}{k_v} + \frac{C}{k_a} \quad \begin{cases} A = 10 \\ B = 60 \\ C = 60 \end{cases}$$

$$k_p = \lim_{s \rightarrow 0} G(s) \rightarrow \infty$$

$$k_v = \lim_{s \rightarrow 0} s G(s) \rightarrow \infty$$

$$k_a = \lim_{s \rightarrow 0} s^2 G(s) = \lim_{s \rightarrow 0} \frac{400(0.3s+1)}{s+1} = 400$$

$$\therefore e_{ss} = 0 + 0 + \frac{60}{400} = 0.15^\circ$$



Q5

$$x_o(t) = \theta_m \sin(\omega t), \quad \omega_{max} = 0.5 \text{ rad/s}, \quad \text{find } |E|_{max}$$

$$e_{max} = 1 \text{ rad/s}^2$$

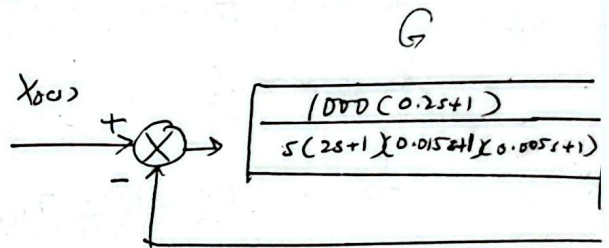
$$\dot{x}_o(t) = \dot{\theta}_o(t) = \theta_m \omega \cos \omega t$$

$$e(t) = \ddot{x}_o(t) = -\theta_m \omega^2 \sin \omega t$$

$$\begin{cases} \theta_m \omega = 0.5 \\ \theta_m \omega^2 = 1 \end{cases}$$

$$\therefore \begin{cases} \omega = 2 \text{ rad/s} \\ \theta_m = 0.25 \text{ rad} \end{cases}$$

$$\theta_o(t) = x_o(t) = 0.25 \sin(2t)$$



$$\frac{1}{1+GH} = \frac{E(s)}{\theta_o(s)} = \frac{1}{1 + \frac{1000(0.25s+1)}{s(2s+1)(0.015s+1)(0.005s+1)}}$$

$$\left| \frac{E(j\omega)}{\theta_o(j\omega)} \right| = \frac{\omega \sqrt{(2\omega)^2 + 1} \sqrt{(0.015\omega)^2 + 1} \sqrt{(0.005\omega)^2 + 1}}{\sqrt{1 + \left(\frac{1000(0.25\omega - 1)}{\omega} \right)^2}}$$

$$\theta_o(t)_{max} = 0.25 \quad \omega(t) = \theta_m \omega \cos \omega t$$

$$\sin \omega t = 1 \quad \omega(t)_{max} = 0.25$$

$$\omega(t) = 2 \text{ rad/s}$$

$$\therefore |E(j\omega)| = 2.08 \times 10^{-3} \text{ rad}$$

Q3 (1) $X_0(s) = 1(s)$, $e_{ss} = ?$

$$G_a = \frac{G}{1 + GH} \quad , \quad H = K_h$$

$$= \frac{\frac{K}{Js + F}}{1 + \frac{K \cdot K_h}{Js + F}}$$

$$= \frac{K}{(Js + F) + K \cdot K_h}$$

$$\frac{E(s)}{X_0(s)} = \frac{1}{1 + (\frac{1}{s} G_a) H'} \quad , \quad H' = 1$$

$$= \frac{(Js + F) + K \cdot K_h}{\cancel{\frac{1}{s}} [(Js + F) + K \cdot K_h]}$$

$$= \frac{s(Js + F + K K_h)}{Js^2 + (F + K K_h)s + K}$$

$\therefore X_0(s) = 1(s)$, $\therefore X_0(s) = \frac{1}{s}$

$$e_{ss}(\infty) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \left(\frac{E(s)}{X_0(s)} \right) \cdot X_0(s) = 0$$

(2) $X_0(s) = \frac{1}{s^2}$ (单位斜坡)

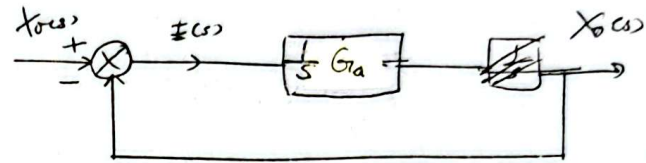
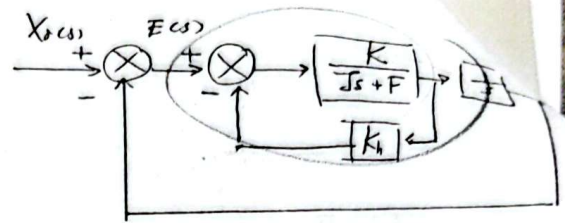
$$X_0(s) = \frac{1}{s^2}$$

$$e_{ss}(\infty) = \lim_{s \rightarrow 0} s \frac{E(s)}{X_0(s)} X_0(s) = \frac{F + K K_h}{K} = \frac{F}{K} + K_h$$

(3) K_h 对 K , e_{ss} 影响, $\therefore e_{ss} = e_{ss1} + e_{ss2}$

K_h 对 e_{ss1} 无影响,

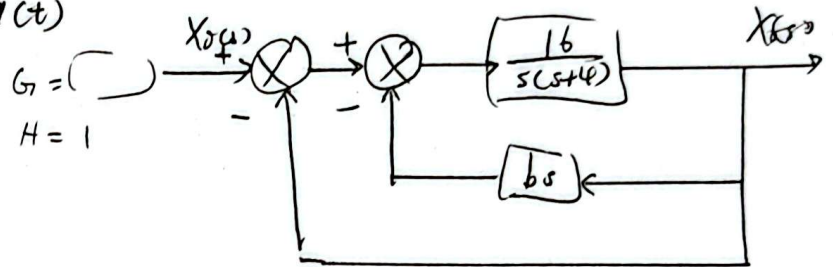
$K_h \nearrow e_{ss2} \nearrow$, $K \searrow e_{ss2} \searrow$ ($\therefore e_{ss}$ 与 K_h 正相关)



c) $b=0$, $x_1(t) = t \cdot 1(t)$

$$\frac{E(s)}{X_1(s)} = \frac{1}{1 + G_1 H}$$

$$= \frac{s(s+4)}{s(s+4) + 16}$$



$\therefore X_1(s) = t \cdot 1(t)$, $X_{1cc}(s) = \frac{1}{s^2}$

~~$\lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \left(\frac{s^2(s+4)}{s(s+4) + 16} \right)$~~

$\lim_{s \rightarrow 0} s \frac{E(s)}{X_{1cc}(s)} = \lim_{s \rightarrow 0} \frac{s(s+4)}{s[s(s+4) + 16]} = \frac{4}{16} = 0.25$

c) $b=0.15$,

$G_1 = \frac{G}{1 + G_1 H}$, $H = bs = 0.15s$

$\therefore = \frac{\frac{16}{s(s+4)}}{1 + \frac{16}{s(s+4)}(0.15s)}$

$\therefore \frac{E(s)}{X_1(s)} = \frac{s(s+6.4)}{s(s+6.4) + 16}$

$\therefore X_{1cc}(s) = \frac{1}{s^2}$

~~$\lim_{s \rightarrow 0} s E(s)$~~ $\lim_{s \rightarrow 0} s \frac{E(s)}{X_{1cc}(s)} = \lim_{s \rightarrow 0} \frac{s+6.4}{s[s(s+6.4) + 16]} = 0.4$